
PRE-TRAINING IN REINFORCEMENT LEARNING: DATA, CURIOSITY AND STRUCTURE

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ABSTRACT

Recent progress in Reinforcement Learning (RL) has been driven by the ability to learn from label-free data. As it becomes easier to handle more data computationally, understanding the generative process of an agent’s interactions with an environment paves the way for the efficient use and deployment of data. With Large Language Models becoming popular, methods of unsupervised learning and self-supervised learning have exhibited their impact on RL. This article is an overview of some emerging research trends in RL. It is a deep dive into selected representative methods of learning from data under a unified framework of core objectives. With the help of this framework, this article provides insights into what types of motivation drive data-driven learning in Deep RL and highlights future directions.

1 Introduction

Label-free learning has enabled recent progress in learning from large amounts of data. Advancements in Large Language Models showcase the effectiveness of learning the generative process from huge amounts of text data. This also pushed learning without labels or learning from auxiliary labels in some of the sub-fields of Reinforcement Learning (Park et al. [2024], Rudolph et al. [2024], Li et al. [2023], Ma et al. [2022], etc.). This article provides a focused study of different works unified by data-driven objectives. It filters the entire discussion of using data into two different perspectives. First is learning driven by curiosity to explore the dataset, and the second is to learn the underlying core information.

This framework is motivated by works that focus on data that is invariant to task-relevant information (Zisselman et al. [2023], Wilcoxson et al. [2024], Rudolph et al. [2024], Laskin et al. [2020], etc.). The article discusses the strengths and key contributions of these works. This discussion highlights the type of underlying structure that exists in the data, what are the ways to learn about this structure, and how to adapt these insights to solve downstream tasks with minimal intervention. The aim is to give an integrated understanding of these works while providing insights that pave the way for future research.

Prevailing practices demonstrate that the availability of large data has made its use and scientific consideration inevitable. But not all data is good. A lot of data used for training RL agents comes from unknown tasks, suboptimal, and incomplete interactions (Hansen-Estruch et al. [2022], Park et al. [2024], Li et al. [2023]). Nevertheless, methods to pre-train using this data are attractive because solving new tasks from scratch is not efficient. Utilizing data is beneficial in cases where interacting with the environment during deployment is not possible.

Most of the papers covered under the discussion presented here are works in unsupervised and self-supervised pre-training. Some works that lay the foundation of such approaches have also been described in the relevant sections. This analysis has been divided into four main sections. Each section covers a pre-training objective that is motivated by the two main factors described above. Most recent works cover these two motivations independently. This article does not propose a novel method to solve any problem, but offers an organized discussion.

The focus for each section is as follows: exploring datasets for solving downstream tasks with minimal or no adaptation, learning representations that capture structure in environmental dynamics, learning representations that capture similar functional behavior within an environment, and learning representations that capture structure between similar and

dissimilar behaviors. The objectives include maximizing entropy, finding temporal structure, learning successor features, learning with contrastive objectives, and bisimulation metrics. These methods provide overlapping answers to some important research questions. These include determining the right type of representations to learn from data and understanding the trade-offs of different pre-training objectives for efficient downstream adaptation.

2 Problem Statement

All problem settings presented operate in an MDP (Markov-decision process) framework or some extension of this framework. The environment is described as an MDP M with state space $\mathcal{S}$, action space $\mathcal{A}$, transition dynamics P , and discount factor γ . A policy $\pi(a|s)$ describes what action to take given a state s . Each task is defined via a reward function r that maps state-action pairs to real-valued rewards. In some cases, an offline dataset of trajectories is available. This dataset, denoted as $\mathcal{D}$, consists of N trajectories, where each trajectory τ_i is a sequence of tuples $(s_t^i, a_t^i, r_t^i, s_{t+1}^i)$ for $t = 0$ to T_i , with T_i being the horizon length of the trajectory. These trajectories can be collected using any policy, whether optimal or suboptimal, and may involve unknown reward functions.

3 Methods

This section focuses on four different approaches for representation learning in RL. These representations allow for few-shot adaptations to downstream tasks. The section describes the different pre-training objectives, adaptation or fine-tuning strategies for new tasks, implementation choices, and open questions. The aim is to cover the strengths and impact of these contributions.

3.1 Exploration-based Representations

Incorporating exploration as an objective can be used to learn diverse skills from data. Exploration-based works focus on invariance to task rewards. This can be done for either the offline data available or for gathering new data through online interactions. Exploring a domain is harder than training a policy to maximize reward in an environment (Zisselman et al. [2023]). A pre-training exploratory policy can be used for zero-shot RL (Zisselman et al. [2023]). This can be done by training an exploratory policy that maximizes the entropy of a marginal state visitation distribution for a finite horizon T . This distribution is defined as:

$$d_{T,\pi}(s) = \frac{1}{T} \sum_{t=0}^T d_{t,\pi}(s), \quad (1)$$

where $d_{t,\pi}(s)$ denotes the t -step state distribution when following policy π .

Intuitively, the entropy objective helps move towards the states that the policy has not encountered in the episode. But learning an exploratory policy is not enough, and disentangling exploration from underlying dynamics is important. One way to do this is to train an ensemble of policies and use the exploratory policy to find actions in the environment when the ensemble fails to agree on an action (Zisselman et al. [2023]). The objective maximized over a horizon T , becomes:

$$R_T(\pi) = \mathbb{E}_{\mathcal{M} \sim \hat{P}(\mathcal{M})} [H(d_{T,\pi}(s))] \quad (2)$$

$$= \mathbb{E}_{\mathcal{M} \sim \hat{P}(\mathcal{M})} \left[H \left(\frac{1}{T} \sum_{t=0}^T d_{t,\pi}(s) \right) \right]. \quad (3)$$

The expectation is over different MDPs sampled from a distribution. This objective learns a policy that equally visits all the states during the episodes (Zisselman et al. [2023]). Since sampling from the state-visitation distribution is not trivial, entropy can be computed by a particle-based k-nearest neighbor approximation (Zisselman et al. [2023], Yarats et al. [2021]). For practical implementation, (Zisselman et al. [2023]) treats the objective as a reward to be optimized for by training an agent with PPO.

Evidently, the Max-Entropy trained policy shows a small generalization gap on the unknown tasks for effective zero-shot generalization. In Zero-shot generalization, a policy is trained on a set of tasks and tested on unknown tasks. The generalization quality of this policy is tested by training agents on 200, 500, and 1000 of Progen’s Maze, Jumper, and Miner (Zisselman et al. [2023]). Such exploration only learns reward invariance. A promising direction is how well such a policy performs when combined with tasks that require invariance to environment features. Another is how to use these environment interactions more efficiently.

A sample-efficient approach with a similar motivation can be used for representation learning. Learn a visual representation without task rewards in a self-supervised way. The pre-training phase has two parts. First, learn a set of prototypical embeddings that form the fundamental basis of a low-dimensional latent space. Second, use these prototypes to maximize an entropy-based intrinsic reward, encouraging exploration of the environment. The intrinsic reward is similar to the one in Zisselman et al. [2023], but instead of states, it uses encoded observations. Yarats et al. [2021] shows the performance of such an approach on downstream tasks, and the intrinsic reward is appended with the task reward for DeepMind Control using SAC.

Both methods assume interaction with the environment in the pre-training phase, with no fine-tuning for downstream adaptations. But diverse behavior from offline data can also be learnt in a task-agnostic way. The idea is to optimize for diverse temporal structure in the pre-training phase. Learn a state representation ϕ that captures the temporal distance using unlabeled trajectories by using the equivalence between temporal distance and the optimal goal-conditioned value function. This representation is used to learn a policy that spans the latent space and captures diverse skills in the offline data (Park et al. [2024]). Few-shot adaptation to the downstream can be achieved by learning the task-dependent latent variable. Box 1. summarizes the pre-training objectives that focus on capturing diversity in data and respective methods for downstream evaluation.

Box 1. Discover Diversity

Pre-Training Objective

- Zisselman et al. [2023]

Entropy:

$$\hat{H}^{k,T}(d_{T,\pi}) \approx \frac{1}{T} \sum_{t=1}^T \log \|s_t - s_t^{k\text{-NN}}\|_2$$

- (i) Train MaxEnt Policy:

$$r_{\text{aux}} = \frac{1}{T} \sum_{t=1}^T \log \|s_t - s_t^{k\text{-NN}}\|_2$$

- (ii) Train ensemble of policies $\pi_r^1, \dots, \pi_r^m$ using r_{ext}

Downstream Evaluation

- (i) Sample actions from the m policies.
If they disagree, use the exploratory policy.

- Park et al. [2024]

- (i) Learn ϕ using expectile loss:

$$\mathbb{E} [L^2(-\mathbf{1}(s \neq g) - \gamma \|\bar{\phi}(s') - \bar{\phi}(g)\| + \|\phi(s) - \phi(g)\|)]$$

- (ii) Train policy with reward:

$$r(s, z, s') = \langle \phi(s') - \phi(s), z \rangle$$

where z is uniformly sampled from unit sphere in $\mathcal{Z}$

- (i) Zero-shot RL:

$$\hat{r} = \langle \tilde{\phi}(s, a, s'), z \rangle$$

$$z^* = \arg \min_{z \in \mathcal{Z}} \mathbb{E}_D \left[(r(s, a, s') - \hat{r})^2 \right]$$

- (ii) Offline Goal-Conditioned RL:

$$z^* = \frac{\phi(g) - \phi(s)}{\|\phi(g) - \phi(s)\|}$$

Curiosity-driven objectives can be used to improve online data collection and online exploration for new tasks. Such objectives are effective in solving sparse reward tasks at test time. The unlabeled data can be labeled with optimistic rewards with RND (Burda et al. [2018]) to guide exploration to unknown states. An RL agent trained with this data can be used to collect data through online interactions and update the reward model from this online experience. This offers an iterative approach for fine-tuning the agent with updated rewards. In Li et al. [2023], the evaluation for this method showed rapid exploration in challenging sparse-reward domains such as the AntMaze domain, Adroit hand manipulation, and a visually simulated robotic manipulation domain.

Such an objective built upon offline data can be used to improve online exploration. The key idea of this is to extract the low-level skills using a variational encoder. The posterior is conditioned by a prior to stay close to the offline dataset. Learn a low-level policy to take actions around the dataset trajectories and learn a high-level policy through online interactions. This method is effective for tasks that need exploration to be solved (Wilcoxson et al. [2024]).

Both these approaches implicitly capture invariance in the environment dynamics while using unlabeled data to guide exploration. Box 2 summarizes the objectives for using prior data and online exploration for these works.

Box 2. Explore Online

Using Prior Data

- Li et al. [2023]

- (i) Train with optimistic rewards:

$$r_{\text{aux}}(s, a) = r_{\theta}(s, a) + \frac{1}{L} \|f_{\phi}(s, a) - \bar{f}(s, a)\|_2^2$$

- Wilcoxson et al. [2024]

- (i) Low-level Policy: $\mathcal{L}_{\theta}(\tau) = \beta D_{\text{KL}}(f_{\theta}(z | \tau) \| p_{\theta}(z | s_0)) - \mathbb{E}_{z \sim f_{\theta}(z | \tau)} \left[\sum_{h=0}^{H-1} \log \pi_{\theta}(a_h | s_h, z) \right]$

Online Exploration

- (i) Online Exploration and Update Reward Model:

$$L(\theta) = \mathbb{E}_{(s,a,r) \sim \mathcal{D}} \left[(r_{\theta}(s, a) - r)^2 \right]$$

- (i) Collect trajectory data τ_i using the low-level policy.

- (ii) Psuedo-Label: $z_i \sim f_{\theta}(z | \tau_i^{[T]}), r_i = r_{\text{aux}}(s_i, z_i)$

- (iii) High-level policy: $\pi_{\psi}(z | s)$

3.2 Structured Representations

Successor Features present a way to learn representations that explicitly exploit structure in the environment. The structure across different tasks remains the same when the environment dynamics are the same. This section focuses on one way to learn such invariance.

3.2.1 Successor Features

Successor Features were introduced to do transfer in RL by exploiting the underlying structure in dynamics to perform well on new tasks (Barreto et al. [2017]). If the expected one-step reward associated with transition (s, a, s') is decomposed into features represented by ϕ and weights w , such a representation can be used to compute the Q -value function. This Q -value function can be decomposed into the task-relevant information and cumulant ψ , which is called Successor Features. Once Successor Features are learnt for different tasks, Generalized Policy Iteration is used to learn a policy that does as good as any other policy on this new task (Barreto et al. [2017]).

One of the issues with an earlier application of this idea is that with SF and GPI, no task-specific information is shared to learn policies (Borsa et al. [2018]). SF&GPI can be combined with the UVFAs (Schaul et al. [2015]) to learn the value functions over the task encodings and learn a span over the policies. While this offers good generalization, the performance of the policies on a new task depends on how the policies are sampled. An alternative approach for learning shared knowledge across different tasks is using an attention head to learn successor features (Carvalho et al. [2023a]). A more robust way to generate approximations of successor features is to learn a distribution of the returns (Carvalho et al. [2023b]). Finally, such structural decomposition can be used to learn to adapt to tasks with non-Markovian rewards. The successor representations capture the expected steps to reach a state from the current state for the first time (Moskovitz et al. [2021]).

3.2.2 Successor Measure

The successor feature decomposition can be extended to state-visitation distributions. Such decompositions learn the structure of the solutions to tasks. The idea is to decompose the Q -function into the successor measure and the reward. The successor measure encodes the probability of going to the next state under the policy (Touati and Ollivier [2021]). The entire family of policies can be parametrized with z , and the Q -function is used to learn the optimal

policies. Learning the successor measure representation for all z allows for adaptation to any reward function. For each z , find $d \times (S \times A)$ -matrices F_z and B such that the matrix M_z^π is $M_z^\pi = F_z^\top B$. Then, for a reward function r , the action-value function is given by $Q_{z,r}^\pi = F_z^\top B r$ (Touati and Ollivier [2021]).

Learning the span of policies does not have to be restricted to optimal policies. Successor Measure can be used to learn representations in the entire space of policies. Following from the Bellman Flow equations of successor measure, the solutions to these exist if it is an affine combination. For practical implementation, this constrained optimization problem can be transformed into a single-player game in Agarwal et al. [2024]. Boxes 3 and 4 describe the definitions.

Box 3 : Successor Features (Barreto et al. [2017])

Reward Function :

$$r(s, a, s') = \phi(s, a, s')^\top w \quad (4)$$

where $\phi(s, a, s') \in \mathbb{R}^d$ are the features of (s, a, s') and $w \in \mathbb{R}^d$ are the weights.

Q-Value function :

$$Q^\pi(s, a) = \mathbb{E}_\pi[r_{t+1} + \gamma r_{t+2} + \dots \mid S_t = s, A_t = a] \quad (5)$$

$$= \mathbb{E}_\pi[\phi_{t+1}^\top w + \gamma \phi_{t+2}^\top w + \dots \mid S_t = s, A_t = a] \quad (6)$$

$$= \mathbb{E}_\pi \left[\sum_{i=t}^{\infty} \gamma^{i-t} \phi_{i+1} \mid S_t = s, A_t = a \right]^\top w \quad (7)$$

$$= \psi^\pi(s, a)^\top w \quad (8)$$

Generalized Policy Iteration (Barreto et al. [2017]) : Let $\pi_1, \pi_2, \dots, \pi_n$ be n decision policies, and let $\tilde{Q}^{\pi_1}, \tilde{Q}^{\pi_2}, \dots, \tilde{Q}^{\pi_n}$ be approximations of their respective action-value functions such that

$$|Q^{\pi_i}(s, a) - \tilde{Q}^{\pi_i}(s, a)| \leq \epsilon, \quad \forall s \in \mathcal{S}, a \in \mathcal{A}, i \in \{1, 2, \dots, n\}. \quad (9)$$

Combined policy

$$\pi(s) \in \arg \max_a \max_i \tilde{Q}^{\pi_i}(s, a) \quad (10)$$

Box 4 : Successor Measure (Touati and Ollivier [2021])

$$M^\pi(s, a; s', a') = \sum_{t=0}^{\infty} \gamma^t \Pr((s_t, a_t) = (s', a') \mid s_0 = s, a_0 = a, \pi) \quad (11)$$

$$Q_r^\pi(s, a) = \sum_{s', a'} M^\pi(s, a; s', a') r(s', a') \quad (12)$$

$$Q_r^\pi = M^\pi r \quad (13)$$

where Q_r^π , M^π , and r are vector representations indexed by state-action pairs.

Successor Features and Successor Measure offer a concise way to represent the underlying structure explicitly. Their ability to transfer to downstream tasks relies on how the space of policies is encoded with this information, making it an important area of closer examination.

3.3 Bisimulation-based Representations

Bisimulation metrics capture invariance to task-irrelevant features. They group functionally similar states in a latent space. Capturing the behavioral similarity in an MDP can be useful for generalization over large state spaces (Ferns

et al. [2004]). In theory, Bisimulation defines an equivalence relation over the state-space. Partitioning the state-space under this relation is difficult for practical application. To deal with this, a semi-metric over this relation is defined to capture how similar two states are. This gives a class of bisimulation metrics. Bisimulation metrics are used to learn representations from image observations in reinforcement learning (Rudolph et al. [2024], Hansen-Estruch et al. [2022]). Box 5 provides the definitions for the equivalence relation and its adaptation for a policy. It also describes an optimization objective to learn representations with the help of a pseudo-metric.

Box 5 : Bisimulation in RL

- **Ferns et al. [2004]**

Given an MDP M , an equivalence relation $E \subseteq S \times S$ is a *bisimulation relation* if whenever $(s, t) \in E$, the following properties hold, where S_E is the state space S partitioned into equivalence classes defined by E :

$$\forall a \in A, \quad R(s, a) = R(t, a), \quad (14)$$

$$\forall a \in A, \forall c \in S_E, \quad P(s, a)(c) = P(t, a)(c), \quad \text{where } P(s, a)(c) = \sum_{s' \in c} P(s, a)(s'). \quad (15)$$

Learning such a representation can be difficult for all action sequences. An on-policy variation of it is defined as follows.

- **Pavse et al. [2024]**

Given an MDP M , an equivalence relation $E^\pi \subseteq S \times S$ is a π -*bisimulation relation* if whenever $(s, t) \in E^\pi$, the following properties hold:

$$R_s^\pi = R_t^\pi, \quad (16)$$

$$\forall C \in S_{E^\pi}, \quad P_s^\pi(C) = P_t^\pi(C), \quad (17)$$

Two states $s, t \in S$ are π -*bisimilar* if there exists a π -bisimulation relation E^π such that $(s, t) \in E^\pi$. A distance function $d \in \mathcal{M}$ is a π -*bisimulation metric* if $E_d = \pi$ -bisimulation relation. $F^\pi : \mathcal{M} \rightarrow \mathcal{M}$ by:

$$F^\pi(d)(s, t) = |R_s^\pi - R_t^\pi| + \gamma W_1(d)(P_s^\pi, P_t^\pi), \quad (18)$$

then F^π has a least fixed point $d_\sim^\pi$, and $d_\sim^\pi$ is a π -bisimulation metric.

Putting bisimulation to practical use has come a long way. Issues related to its efficient computation and upper bounds for arbitrary policies proved worthwhile. A sampling-based approximation to the joint distribution of the transition dynamics made it easier to implement for learning approximation to a large-state space (Castro et al. [2021]). A common problem that emerges is that of representation collapse, where two dissimilar states end up with the same representations. One way to mitigate this is to use cosine distance to measure the distance between two latent states (Zang et al. [2022]). Efficiently computing the approximation to the Wasserstein distance is an open challenge.

$$\cos_\phi(x, y) = 1 - \frac{\phi(s)^\top \phi(t)}{\|\phi(s)\| \cdot \|\phi(t)\|}, \quad (19)$$

$$F_{\cos \phi}^\pi(s, t) = |r_s^\pi - r_t^\pi| + \gamma \mathbb{E}_{s' \sim \hat{P}_s^\pi, t' \sim \hat{P}_t^\pi} [\cos_\phi(s', t')]. \quad (20)$$

Aforementioned methods provide objectives to learn effective representations efficiently using bisimulation metrics. The following subsections describe three domains where bisimulation metrics have been adapted for specific problems. More important to this discussion are works on learning action representations and offline data with no action information. In general, these works show the effectiveness of capturing this behavioral similarity in different settings. These adaptations have been successful in solving complex tasks such as robot manipulation (Hansen-Estruch et al. [2022], Shi et al. [2024]).

3.3.1 Goal-conditioned Bisimulation Relations

Policy bisimulation can result in the effective transfer of skills across analogous tasks. Given an MDP M , *goal-conditioned bisimulation relation* $\mathcal{B}$ can be defined (Hansen-Estruch et al. [2022]). For all state - goal pairs $(s, g_s), (t, g_t) \in S \times G$ that are equivalent under $\mathcal{B}$, the following conditions hold:

$$R(s, a, g_s) = R(t, a, g_t), \quad \forall a \in A \quad (21)$$

$$P(G \mid s, a) = P(G \mid t, a), \quad \forall a \in A, G \subseteq S. \quad (22)$$

For practical implementation, an *on-policy* version of this relation gives rise to a paired-state metric:

$$d((s, g_s), (t, g_t)) = |R(s, \pi(s, g_s), g_s) - R(t, \pi(t, g_t), g_t)| \\ + W_2(P(\cdot \mid s, \pi(s, g_s)), P(\cdot \mid t, \pi(t, g_t))), \quad (23)$$

where π is the goal-conditioned policy. The following objective can be used to learn state-goal representations and train an offline goal-conditioned policy. Hansen-Estruch et al. [2022] used a dataset collected using a noisy expert to show skill transfer in manipulation tasks using such a bisim relation.

$$L_\varphi = \left(\|\phi(s, g_s) - \phi(t, g_t)\|_1 - \|R_s - R_t\|_2 - \gamma \|\bar{\phi}(s', g'_s) - \bar{\phi}(t', g'_t)\|_2 \right)^2 \quad (24)$$

A bisimulation objective can be appended with a forward model error-based intrinsic reward to improve exploration. This enhances exploration in the latent space for learning a bisimulation-based representation. This has been empirically shown to learn robust state representations for downstream goal-conditioned tasks (Kemertas and Aumentado-Armstrong [2021]).

3.3.2 Learning from Offline Data

π -bisimulation-based representations can be learnt using an offline dataset. Learning from an offline dataset is notoriously difficult using function approximation. The representations learnt using the bisimulation metric can result in a value function that does not diverge. These representations stabilize TD-learning and are Bellman complete (Pavse et al. [2024]). Pavse et al. [2024] defines a π -bisimilarity kernel $k^{\pi_e} : S \times S \rightarrow \mathbb{R}$ for pairs of state-actions and under π_e :

$$k^{\pi_e}(s, a_s; t, a_t) = k_1(s, a_s; t, a_t) + \gamma k_2(k^{\pi_e})(P^{\pi_e}(\cdot \mid s, a_s), P^{\pi_e}(\cdot \mid t, a_t)), \quad (25)$$

$$(26)$$

Here, k_1 measures short-term similarity based on the rewards received. k_2 measures long-term similarity between probability distributions by evaluating similarity between samples of the distributions according to k^{π_e} .

To use the bisimulation metric for offline datasets, it must learn from an incomplete dataset with missing transitions. While Implicit Q learning can be used to mitigate this problem (Hansen-Estruch et al. [2022]), the error in the estimated value of bisim operator can be reduced by direct use of an expectile operator (Zang et al. [2023]).

$$F^{\pi_\beta} \phi^{\pi_\beta}(s, t) := \arg \min_{\phi^{\pi_\beta}} \mathbb{E}_{a_s \sim \pi_\beta(\cdot \mid s), a_t \sim \pi_\beta(\cdot \mid t)} \left[\tau [\hat{\varepsilon}]_+^2 + (1 - \tau) [-\hat{\varepsilon}]_+^2 \right], \quad (27)$$

$$\hat{\varepsilon} = \mathbb{E}_{s' \sim T^\pi(s), t' \sim T^\pi(t)} \left[|r(s, a_s) - r(t, a_t)| + \gamma \bar{\phi}^{\pi_\beta}(s', t') - \phi^{\pi_\beta}(s, t) \right]. \quad (28)$$

For real-world deployment of bisimulation-based representations, naively training with adversarial states and actions for such cases does not transfer well to the downstream tasks (Yin et al. [2024]). Learn robust representation with perturbed states and goals closer to the negative samples with a contrastive objective.

3.3.3 Action Invariance

Bisimulation-based metrics enable capturing invariance in behavior by learning action representations. Such long-term action representations can be learnt in a self-supervised way using bisimulation (Shi et al. [2024]). The state-conditional action chunk (denoted by c) bisimulation metric is a function $d : \mathcal{S} \times \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$d(c_i, c_j | s_t) = R_{s_t}^{c_i} - R_{s_t}^{c_j} + \gamma W_2(P_{s_t}^{c_i}, P_{s_t}^{c_j}; d_c), \quad (29)$$

where d is a pseudometric and W_2 is the 2nd Wasserstein distance between two distributions. Here, $R_{s_t}^c$ represents the cumulative discounted reward for executing chunk c starting at s_t and $P_{s_t}^c$ represents the distribution of s_{t+k} after executing c from s_t . Such representations are effective for solving complex tasks. This is experimentally shown with 7DOF ARM Control (Shi et al. [2024]). Learning action representations is not restricted to online interactions but can also be achieved from an offline dataset. To stabilize learning close to the behavioral policy and mitigate distribution mismatch, the following objective can be used (Gu et al. [2022]):

$$L(\phi) = \mathbb{E}_{s, a_i, r \sim \mathcal{D}, a_j \sim \mathcal{A}} \left[\left| \|\phi(a_i) - \phi(a_j)\|_1 - \hat{d}(a_i, a_j | s) \right|^2 \right], \quad (30)$$

where

$$\hat{d}(a_i, a_j | s) = |r_i - \hat{R}(s, \phi(a_j))| + \gamma W_2(\hat{P}(\cdot | s, \phi(a_i)), \hat{P}(\cdot | s, \phi(a_j))) + p \cdot \hat{I}_\beta(a_j | s). \quad (31)$$

Here, $d(a_i, a_j | s)$ is an estimate of the true distance between actions conditioned on the same state s . $\hat{R}$ and $\hat{P}$ are the learned reward and transition models, trained separately. $\hat{I}_\beta(a_j | s)$ is a trainable model predicting whether a_j is out-of-distribution.

While the method above explicitly handles the behavioral distribution, a similar objective for single-step action representation, denoted by ϕ , can be learnt independent of data-collecting policy. Such an action-bisimulation metric is defined as:

$$d_{\text{a-bisim}}(s_i, s_j, \phi, \varphi) = (1 - c) \cdot \|\phi(s_i) - \phi(s_j)\|_1 + c \cdot \mathbb{E}_{a \sim \mathcal{U}(\mathcal{A})} \left[W_1(f(\varphi(s_i), a), f(\varphi(s_j), a)) \right], \quad (32)$$

where $c \in [0, 1]$ balances the contributions of the state and action-conditional terms, W_1 denotes the 1st Wasserstein distance and f is the trained forward model. The state representations are learnt by minimizing the L_1 distance between embedded representations $\varphi(s_i)$ and $\varphi(s_j)$ to the action-bisim metric (Rudolph et al. [2024]):

$$L(\mathcal{D}) = \frac{1}{N} \sum_{s_i, s_j \sim \mathcal{D}} \left| \|\varphi(s_i) - \varphi(s_j)\|_1 - d_{\text{a-bisim}}(s_i, s_j, \psi, \varphi) \right|. \quad (33)$$

Bisimulation can effectively capture the similarity between states. It can be adapted for offline datasets, online exploration, and learning action representations.

3.4 Contrastive Learning based Representations

Contrastive Learning objectives have been used for unsupervised RL. It is a framework to learn representations by exploiting the structure between similar and dissimilar pairs of input. This can be achieved in an unsupervised way by performing a dictionary lookup task wherein the positives and negatives represent a set of keys with respect to a query (or an anchor). There are many ways in which such an objective can be captured (Laskin et al. [2020]). One of the most adapted ones is the InfoNCE loss function.

Box 7 : InfoNCE loss function

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\text{sim}(q, k^+)/\tau)}{\sum_{k \in \mathcal{K}} \exp(\text{sim}(q, k)/\tau)} \quad (34)$$

Here, q is the query embedding, k^+ is the positive key embedding, $\mathcal{K}$ is the set of all keys in the mini-batch (positives and negatives), $\text{sim}(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$ denotes the cosine similarity, and τ is the temperature hyperparameter.

Contrastive Learning can be used to train an RL agent over representations learnt from image-based observations. Laskin et al. [2020] uses a modification of the function in Box 7 to train agents for handling both continuous and

discrete action spaces. Contrastive Learning can be used to improve the robustness of bisimulation representation for transfer to downstream tasks. For this, Yin et al. [2024] uses it for perturbing the negative samples. P_{trb} in the following objective, Eq. 38, denotes the learnt perturbation parametrized by θ .

$$\mathcal{L}(\theta) = -\phi(P_{trb}^i(s), P_{trb}^i(g))^\top \phi(s, g)^- \quad (35)$$

Finally, Contrastive Learning is effective for representations over offline image data. Learn representation over states using a goal-conditioned offline pre-training objective as in Ma et al. [2022]. The objective minimizes the distance between the goal-conditioned state-occupancy distribution of the policy and the data distribution. The dual of this objective yields a contrastive RL objective. The objective is:

$$\max_{\pi_T, \phi} \mathbb{E}_{\pi_T} \left[\sum_t \gamma^t r(o; g) \right] - D_{KL}(d_{\pi_T}(o, a_T; g) \| d_D(o, \tilde{a}_T; g)), \quad (36)$$

where $d_{\pi_H}(o, a_H; g)$ is the distribution over observations and actions visited by the policy π_H conditioned on goal g . $d_D(o, \tilde{a}_H; g)$ is the distribution over observations and dummy actions $\tilde{a}_H$ in the dataset D , conditioned on goal g . This method has been effective for zero-shot generalization in goal-conditioned reinforcement learning.

4 Conclusion

This article provides an overview of recent research trends in RL. It looks at different works using unsupervised and self-supervised methods of learning in RL and provides a unification of the core objectives. It covers approaches based on exploration, successor features, bisimulation, and contrastive learning, highlighting the importance of curiosity and structure in RL. This synthesis of reviewed work indicates certain key insights. Curiosity-based approaches capture a notion of diversity and learns to disentangle this diversity for downstream tasks. In general, how exploration is related to representation learning for downstream performance deserves additional study. On the other other, Successor Features and Successor Measures capture structure explicitly. Structure in the environment evidently dictates the structure in the solution space, but this interdependence needs to be looked into further. Bisimulation learns behavioral similarity, but computing the metric using probability distributions is a bottleneck. Contrastive Learning provides a sample-efficient way to learn representations that are close to positive samples and distant from negative samples.

5 Use of Generative AI

LaTeX code for equations has been modified from templates generated with the help of ChatGPT (OpenAI). In addition, it has been used to assist in finding word phrasing. The text was proofread using Grammarly’s basic grammar and style suggestions, without the use of AI-generated content.

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